

Altium Workshop

9/21/26

Introduction

A little bit about us:

Robert

- Engineering Technology undergrad
- Not an officer. Done with school after Christmas
- Mentored in Altium by Tyler R. (LBR)
- Created my first PCB on KiCAD
- I like creative writing and run club

Muhammad

- Computer engineering undergrad
- Mentored by Tyler R & Zey (now at UCSD)
- Don't know KiCAD
- EATs officer, I do micromouse
- Calisthenics & rock climbing

Links

1. PCB Design Courses by Altium | Altium Education
 - Link for student license
 - <https://education.altium.com/>
2. Downloads for PCB Design Products and EDA Software | Altium
 - Link for Altium download
 - <https://www.altium.com/products/downloads>

What is an Altium Workspace?

[Altium 365 Workspace | Altium 365 Technical Documentation](#) (click here for more info)

In Altium, a workspace is a centralized environment for managing, storing, and collaborating on all design data, integrated with Altium Designer and Altium 365.

Think of it like google drive, but instead of docs and spreadsheets, you're collaborating on PCB files

Getting Started

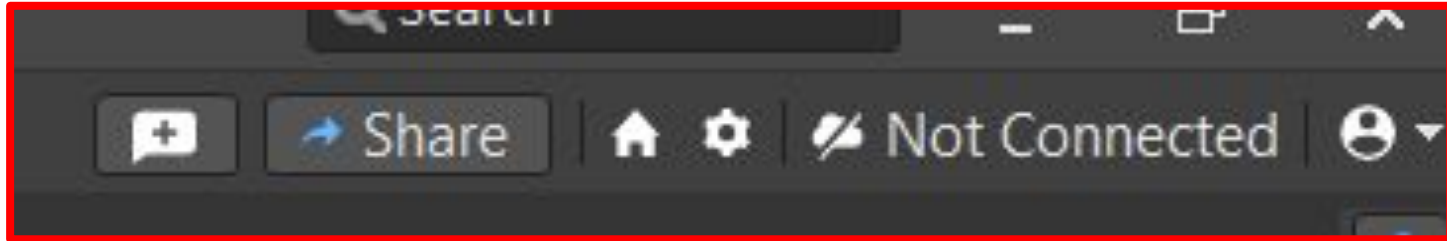
Recap of Session 1: Project and Schematic

Objectives last session

- ❖ Familiarize yourself with Altium workspace and file types
 - Project
 - Schematic
 - PCB
 - Symbol libraries, Footprint libraries
- ❖ Import component models into your workspace
- ❖ Assemble a schematic as instructed by a component datasheet's application page
- ❖ Navigate the schematic document and its tools (nets, power flags, etc.)

First step: Connect your Altium account

Whenever you open altium, check the top right. If not connected, select “not connected” and connect to your account



Make sure you are using the professional license

LDO_Tutorial.PrjPcb - Altium Designer Professional (26.10.1)

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Robert Bagalawis

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Product	Subscription Renew	Seats	License Info	
Altium Designer Professional In Use	3/24/2027	1/1	On-Demand · Single Site · Term-Based · N3VM-4B36	Roam Release
Altium Designer Viewer	—	0/1	On-Demand · Single Site · Viewer · YEFY-LKU9	Use License

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46

Bookmarks



“Projects” panel should appear on screen
If not visible:
view > panels > projects

Shortcut: Bottom right corner should show
“Panels”
If not visible: view > status bar



Welcome, Robert Bagalawis

CERTIFICATES:



0 of 5

[Training Catalog](#) →

ates

Webinar

Filters: All



Blog • Sep 21, 2026 •

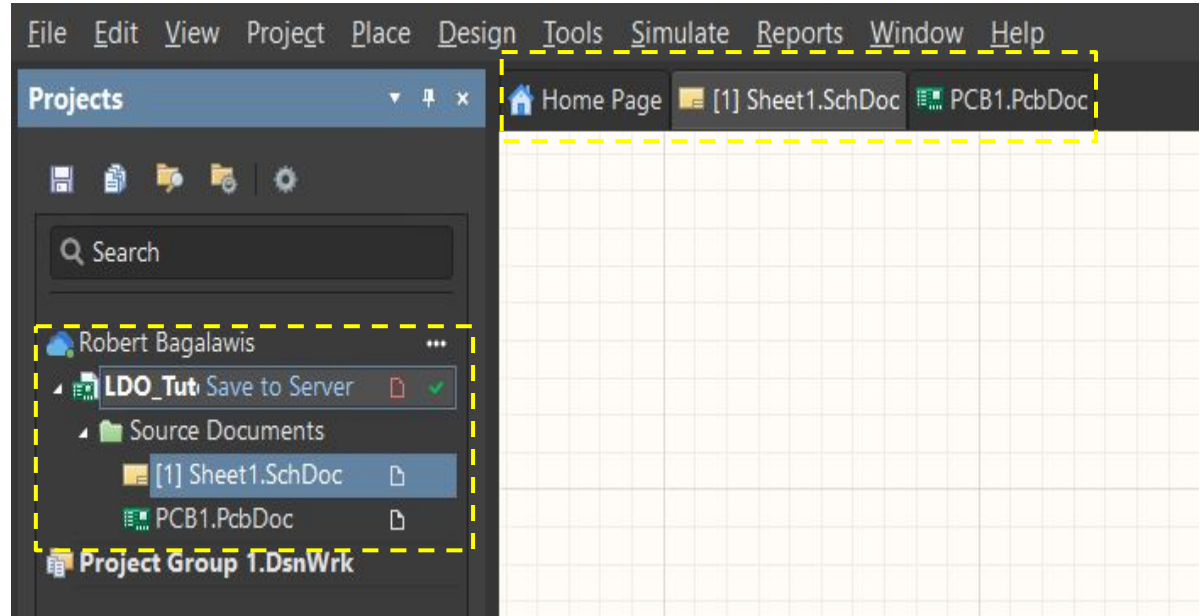
Ultra HDI Is Changing What “Assembly-Friendly” Looks Like

As component pitch decreases, even small manufacturing variations can consume a significant portion of the available design margin. This article explains how tolerance stacking, solder paste deposition, inspection limitations, and rework challenges can affect assembly reliability in Ultra HDI designs.

[Read More](#)[Open in Browser](#)

Create New Project and Files

- File>New>Project
- Inside your project
 - File>New>Sch
 - File>New>PCB



<https://github.com/rbagalawis2001/PCB-Design.git>

Download these three components from my github repo and unzip (extract)



Part Number	Component	Parameters (simple)	Why we're using it
MIC5365-3.3YC5-TR	Low Dropout Voltage Regulator	Input: 2.5V - 5.5V Output: 3.3V	LDO keeps a steady 3.3V output with low noise
691137710002	Screw Terminal	2 position	Wires connections hold firm and are interchangeable
GRT188R61H105KE13D	Ceramic capacitor	1 microfarad	Stores charge and smoothes out voltage ripples

Files

main

Go to file

AltiumTutorialCSULB

- 691137710002.zip
- GRT188R61H105KE13D.zip
- MIC5365_3_3YC5_TR.zip

PCB-Design / AltiumTutorialCSULB

rbagalawis2001 Add files via upload d9f66e1 · 10 minutes ago History

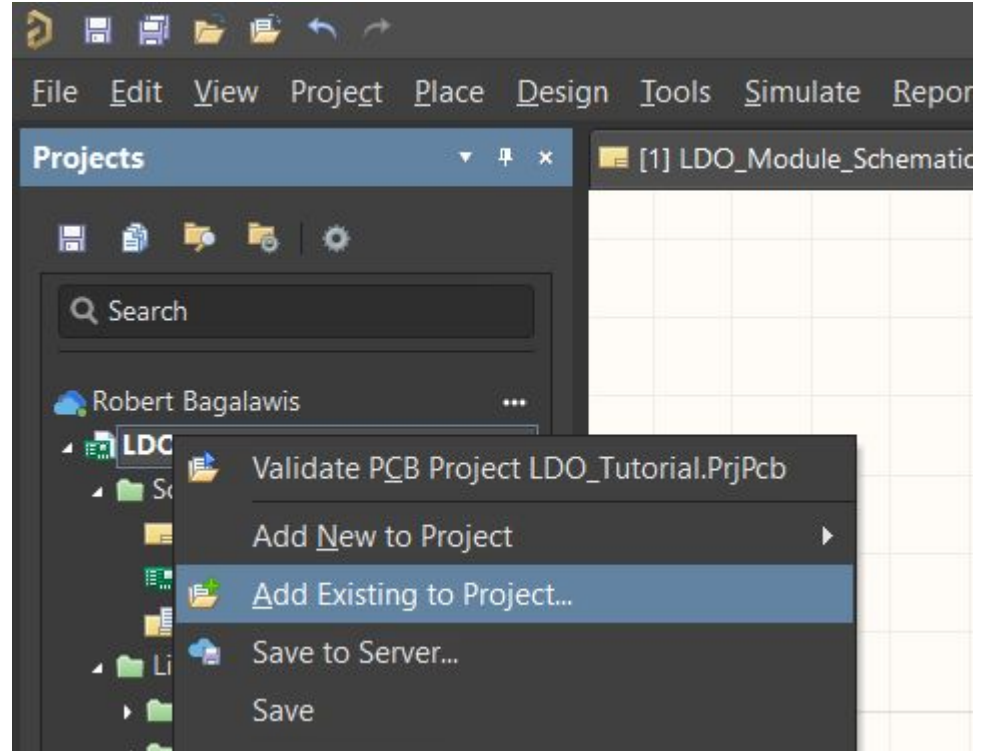
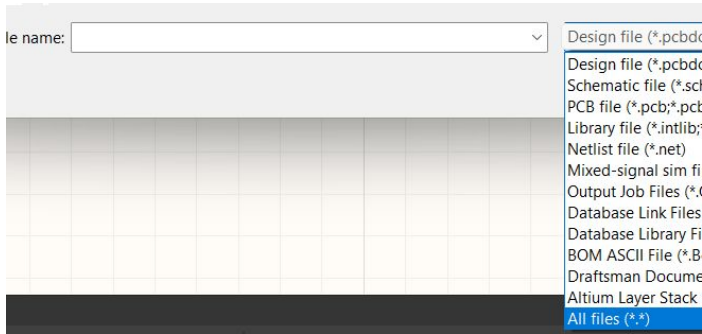
Name	Last commit message	Last commit date
..		
691137710002.zip	Rename 691137710002.zip to AltiumTutorialCSULB/691137710002.zip	4 hours ago
GRT188R61H105KE13D.zip	Add files via upload	10 minutes ago
MIC5365_3_3YC5_TR.zip	Rename MIC5365_3_3YC5_TR.zip to AltiumTutorialCSULB/MIC5365_3_3YC5_TR...	4 hours ago

Importing libraries

Right click on .prjpcb>add existing

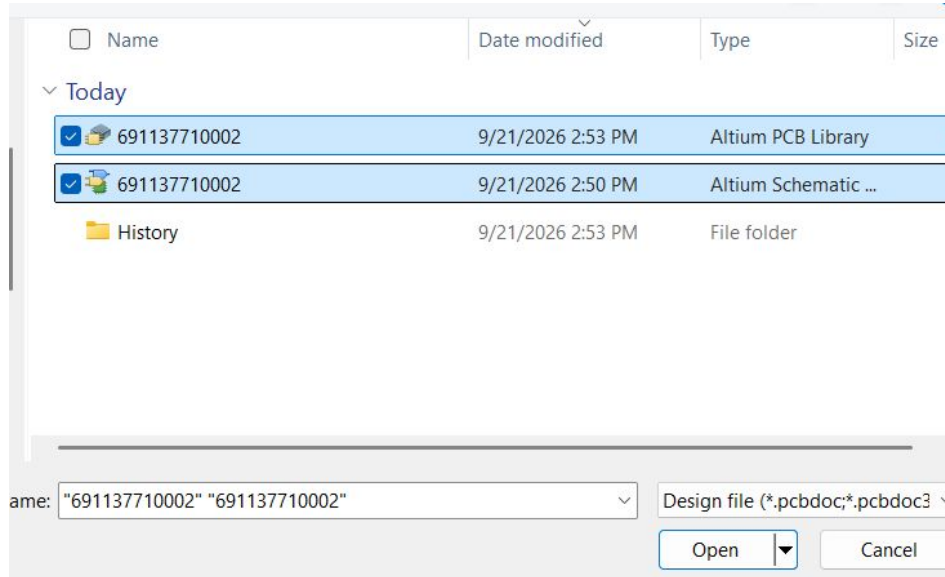
Search for your downloaded files.

May need to select “all files”



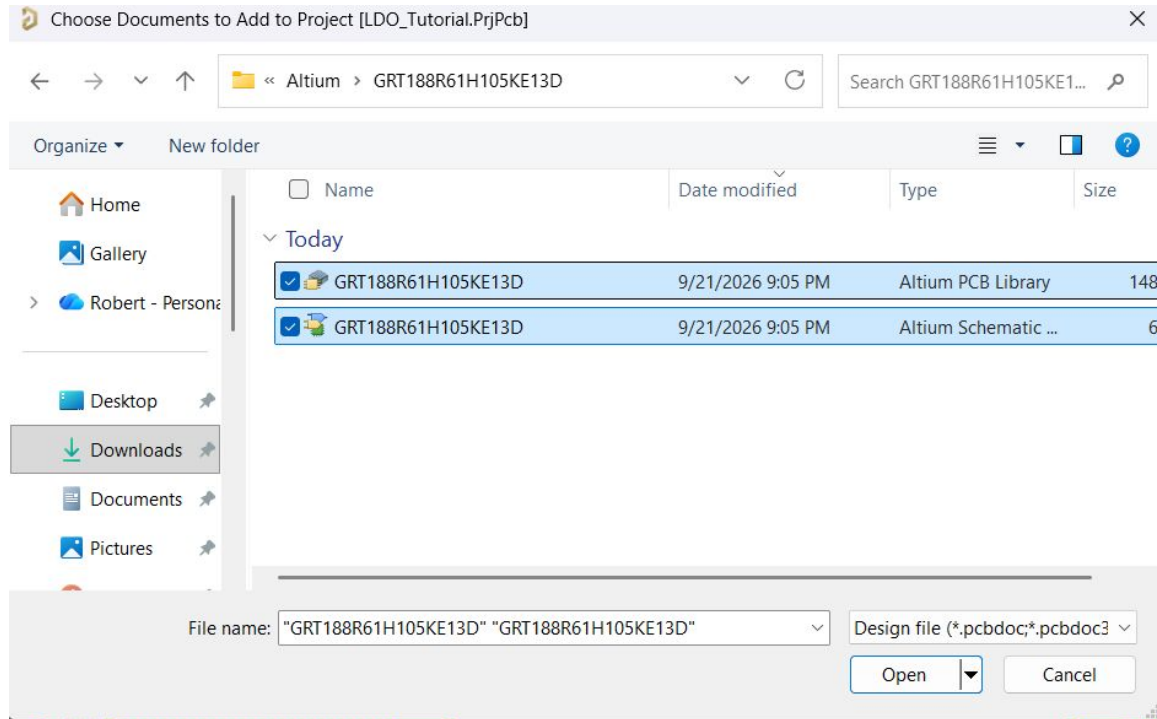
Importing– Screw Terminal

Hold Shift > Select both Schematic Library and PCB Library



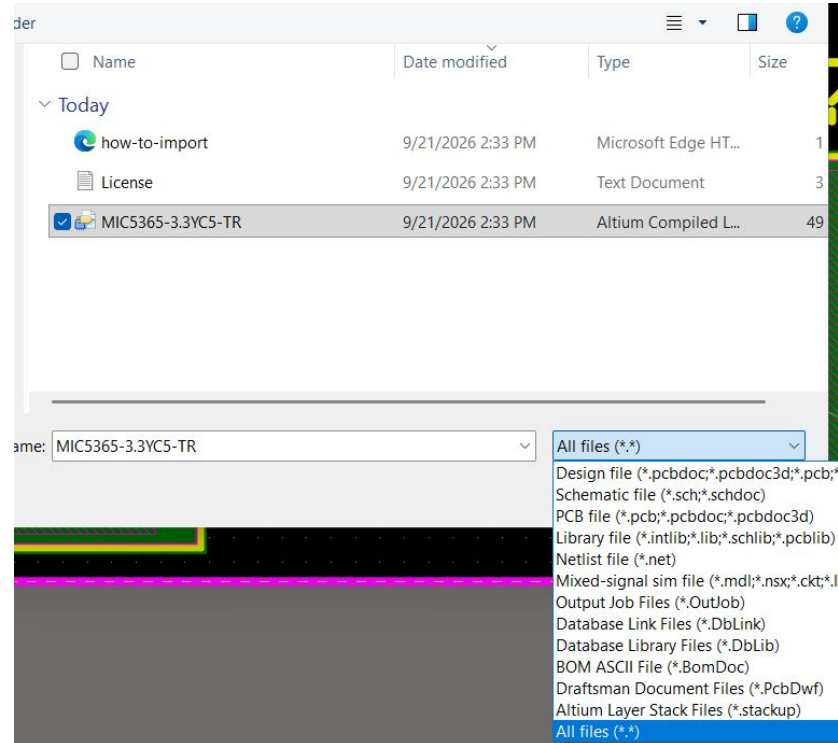
Importing– Ceramic Capacitor

Hold Shift > Select both Schematic Library and PCB Library



Importing— Low Dropout Regulator

You may have to change your file selector to “all files”

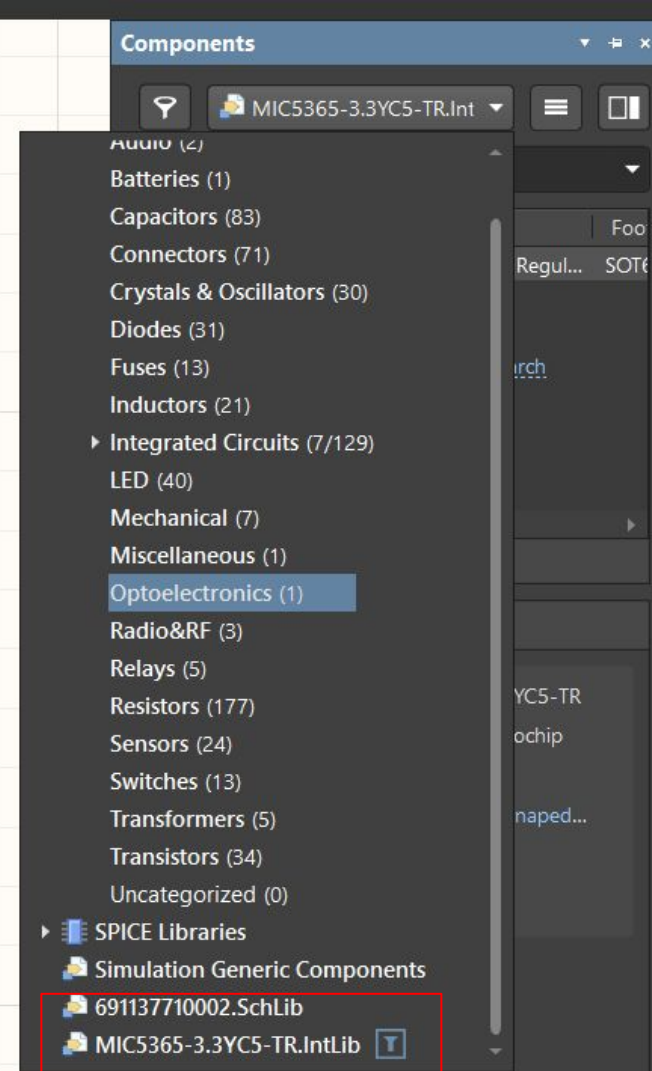
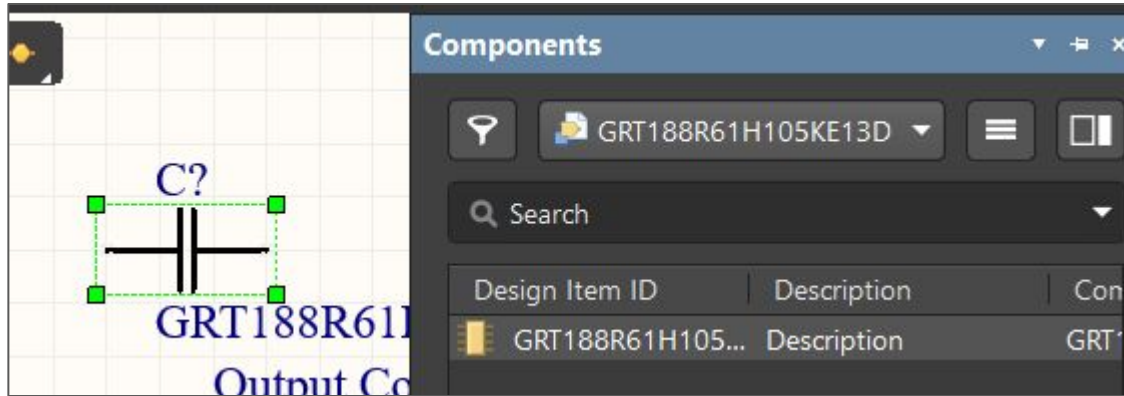


Add imported parts

Components > All > scroll to your imported library

>select the component library----->

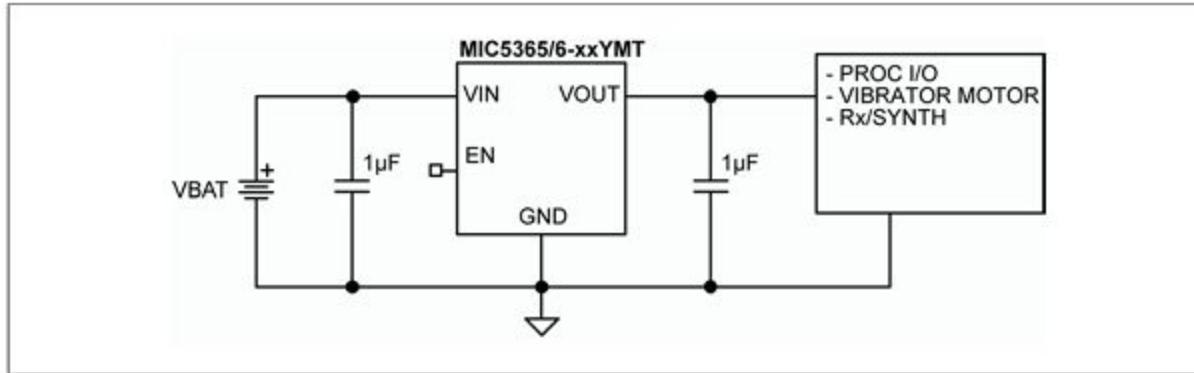
Click on the component and drag it to your schematic



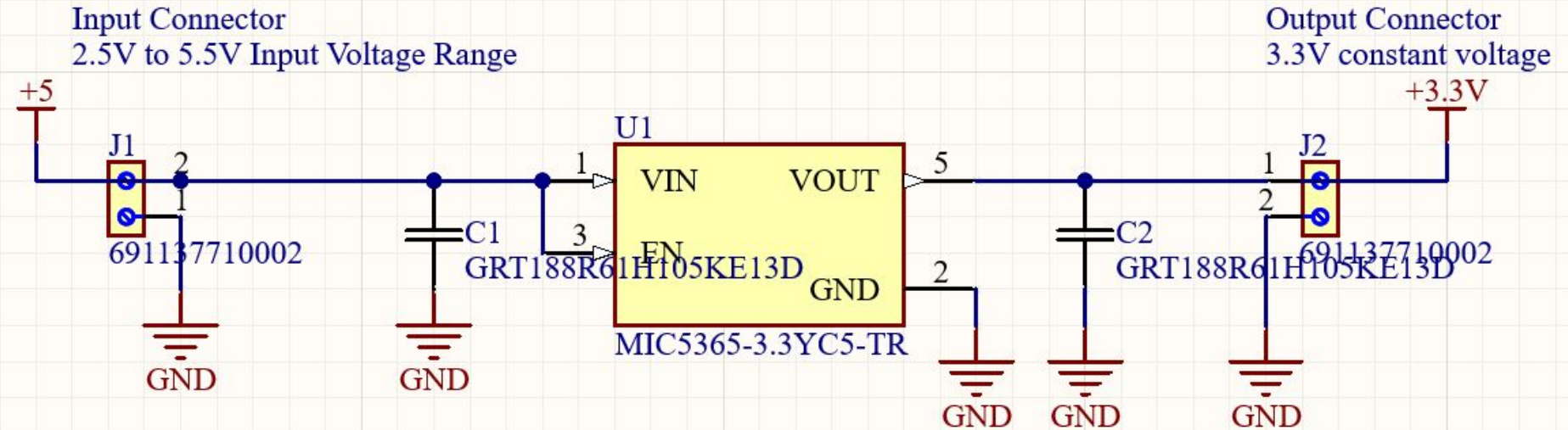
Datasheet Schematic (Application Page)

MIC5365/6

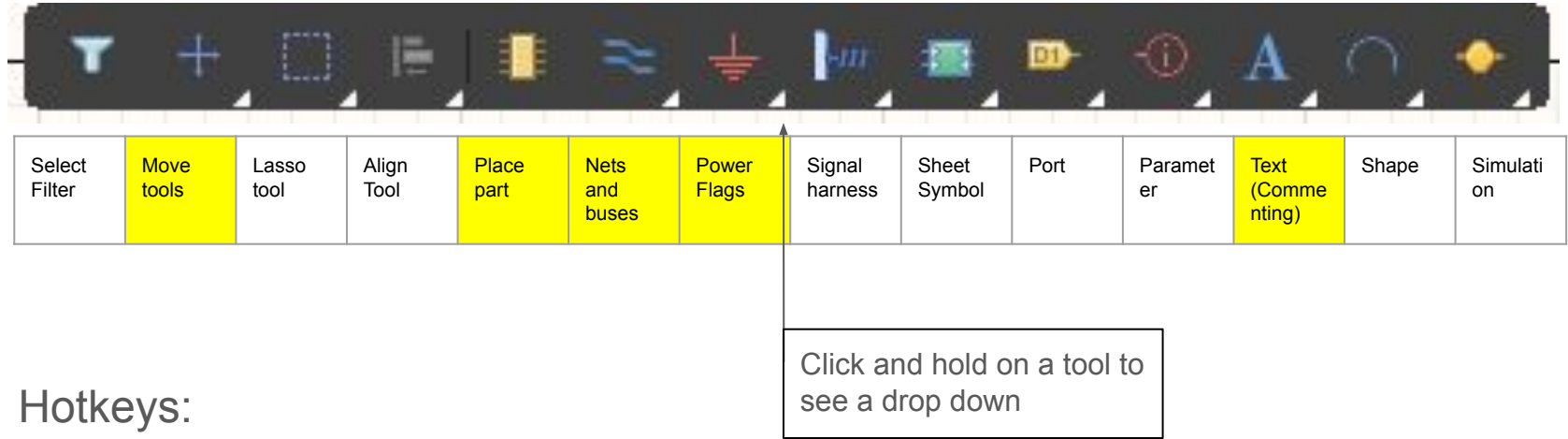
Typical Application Circuit



Exercise: Assemble this



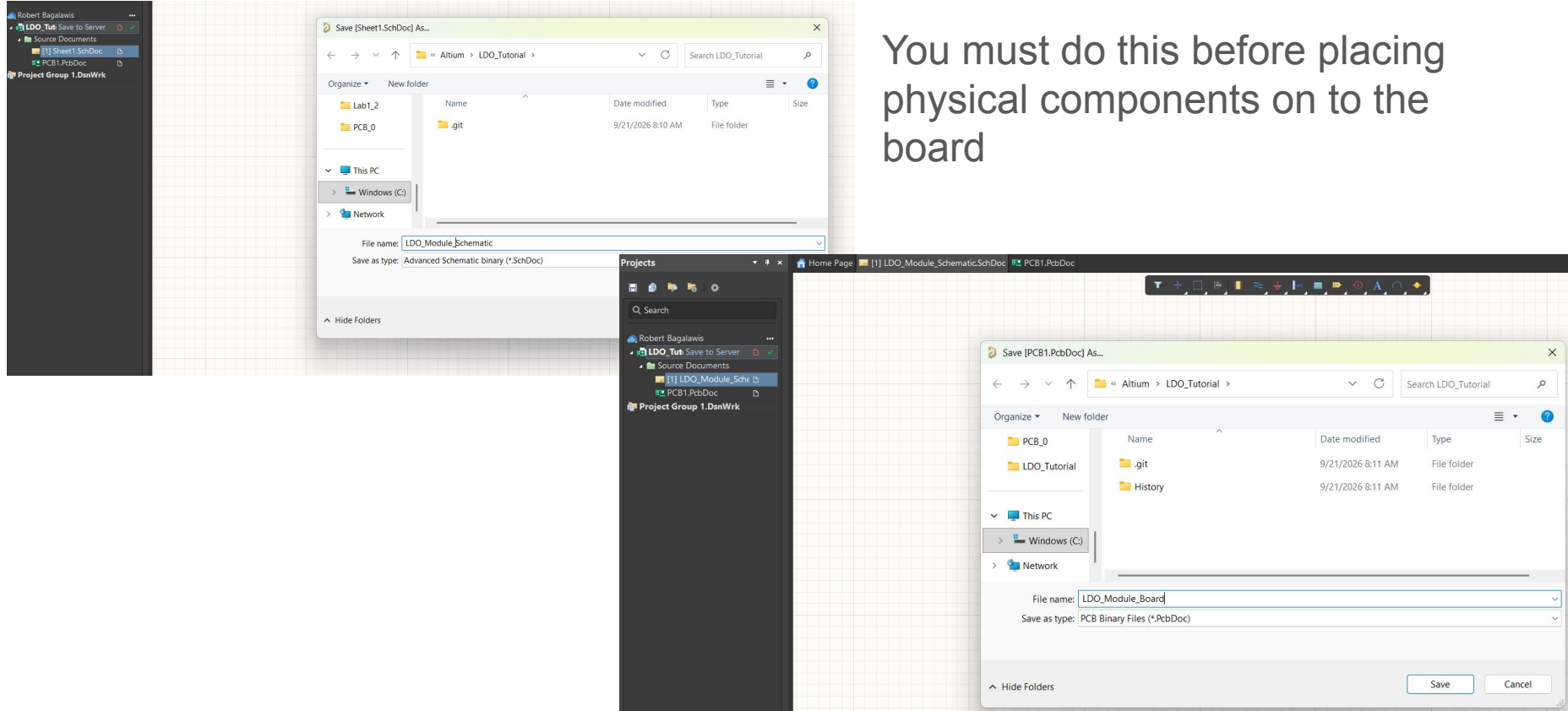
Here are your SCH design tools



Hotkeys:

- CTRL + W: Draw a net (Wire connecting two schematic pins)
- When a component is selected, press spacebar to rotate. You can also rotate components before placing them on the schematic
- Press "M" twice to move a component without moving the net it's attached to

Save to Server



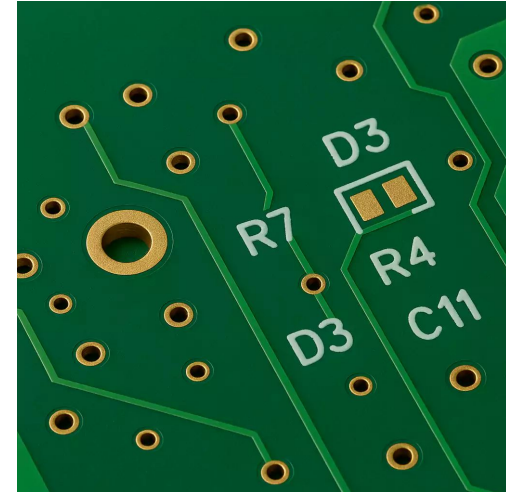
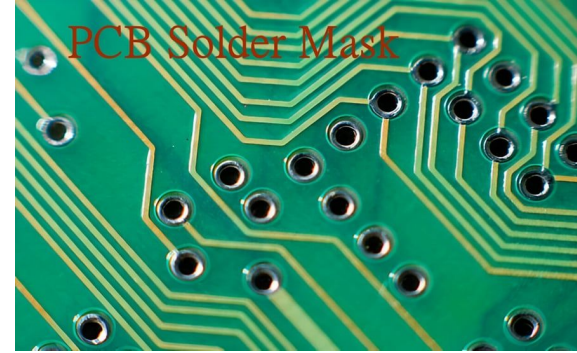
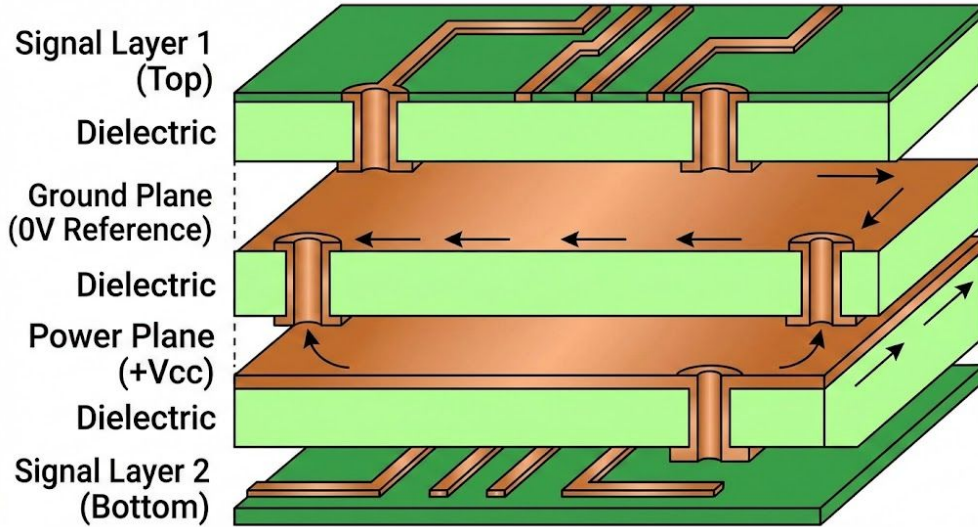
Picking up where we left off

Session 2: Physical board design

Objectives today

- ❖ Place physical component markers (footprints) on the PCB document
 - Understand the relationship of schematics/symbol and PCB/footprint
- ❖ Navigate the PCB document
 - Component rotation and correct orientation
 - Create minimal board outline
- ❖ Navigate the 3D viewer
 - Toggle view angle
 - Understand user interface with your PCB and importance of component placement

PCB layers

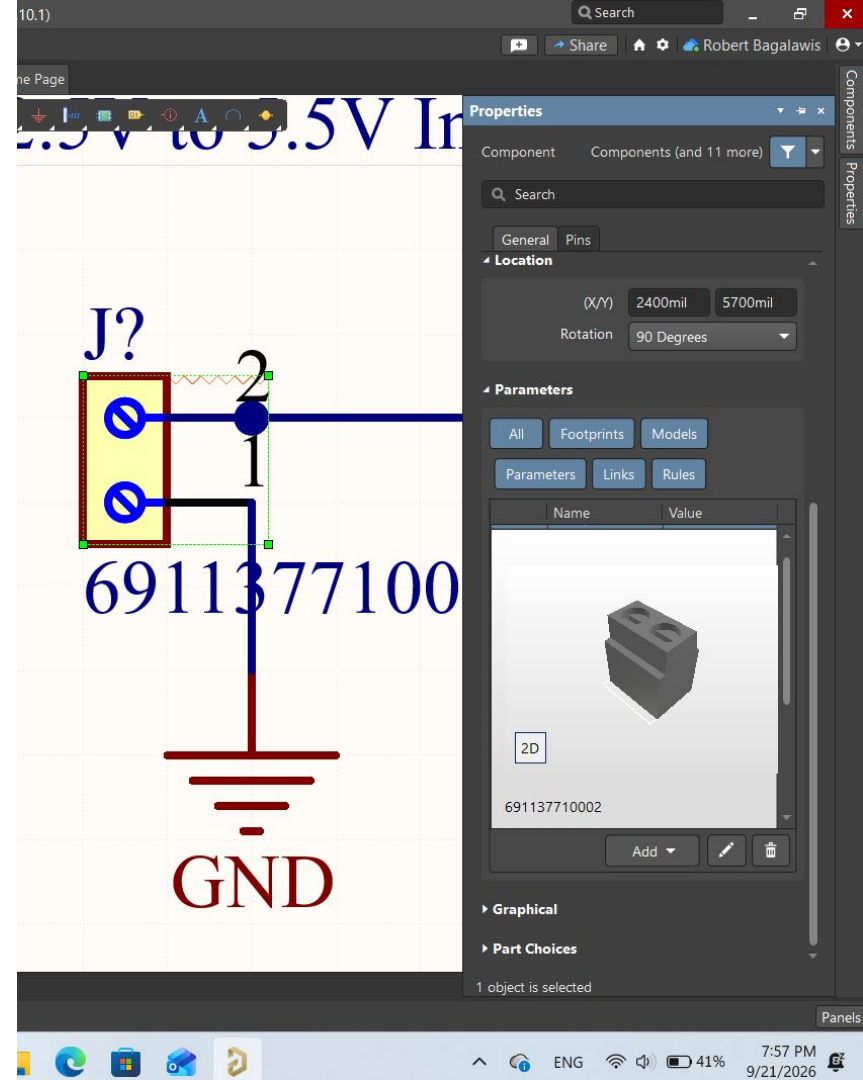


Symbol-Footprint-3DModel Assignment Check

Select your component > Right click > Properties

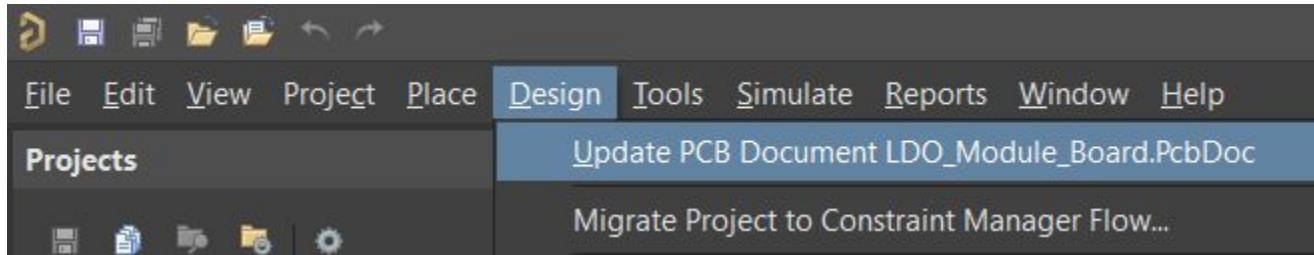
Under Parameters: make sure Footprint is selected (Highlighted in blue) > check to see if footprint and 3D model are visible

If the footprint/3D model are not visible on properties, it will not show on the PCB editor

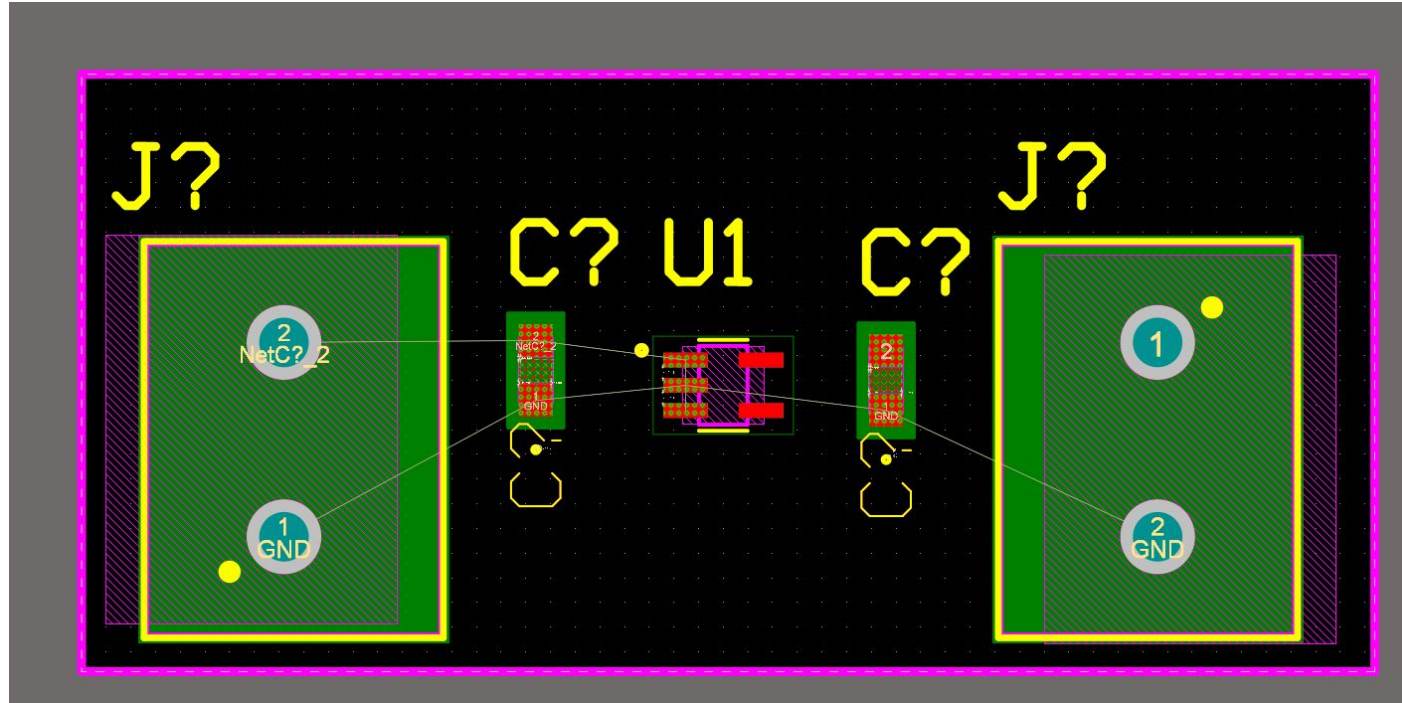


Place physical components into you PCB document

Design > Update PCB

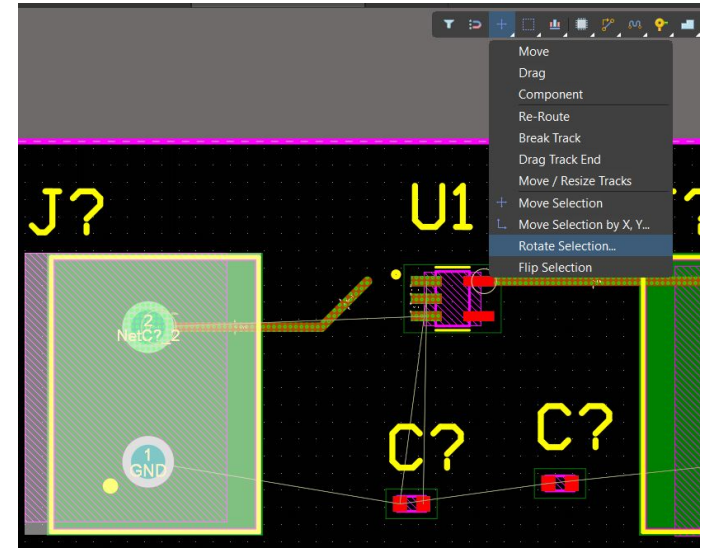


Exercise: Assemble this



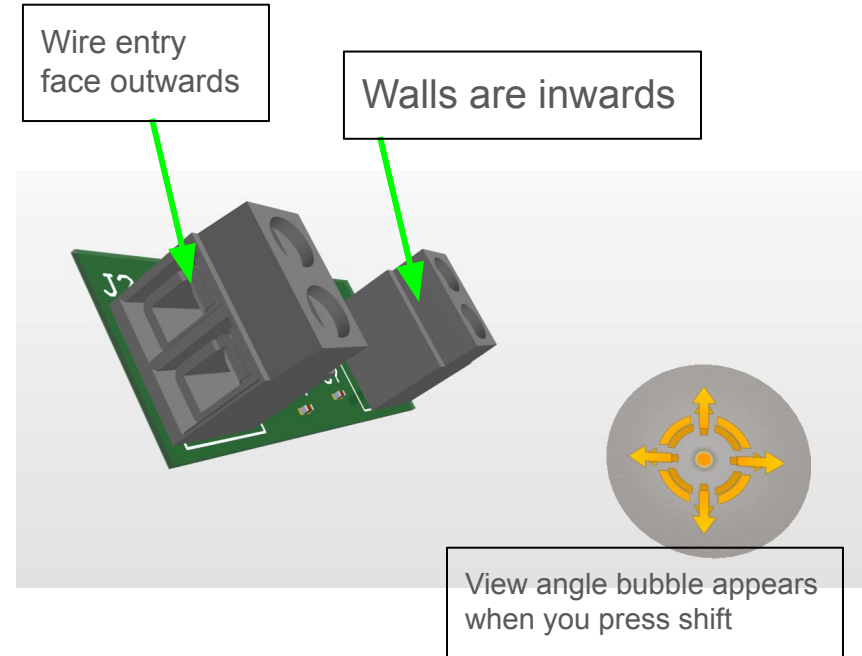
PCB Navigation

- Press “3” for 3D view: Shows STEP file of 3D model components and a preview of the manufactured board
- Press “2” for 2D view: Pin-pin wiring (Traces) in all board layers and electrical terminals (footprints)
- Rotate a component by selecting it > click and hold on the move tool > Rotate selection > input rotation degrees > click on the component again



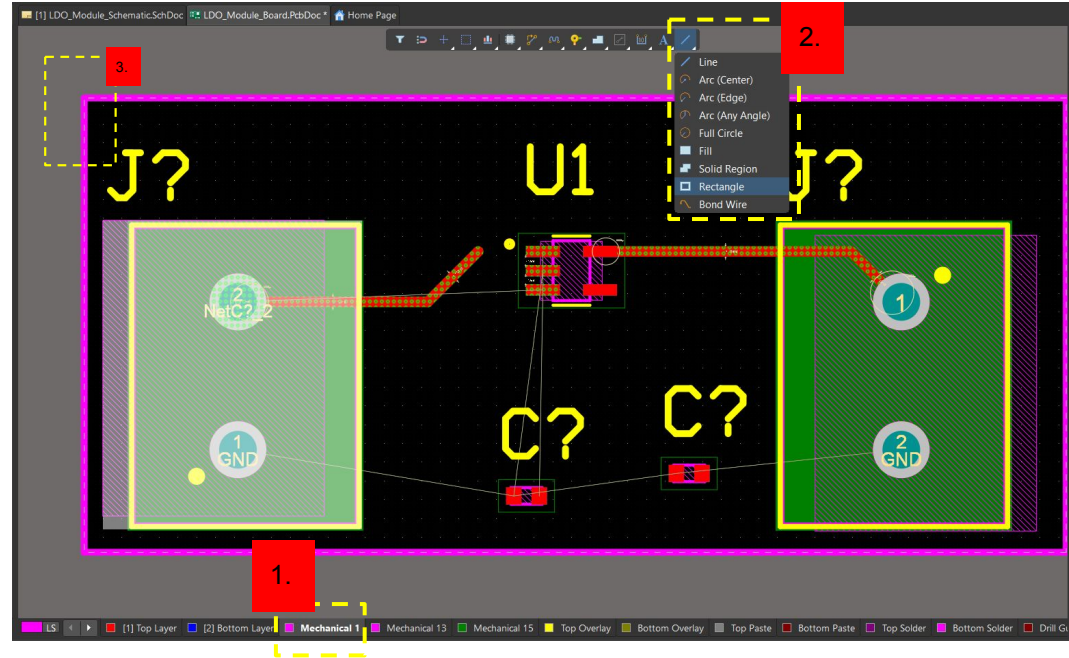
PCB Navigation 3D Viewer

- To verify the orientation of your components, switch to 3D view.
- In this example, we want wire entry points to face outwards
- Press shift and hold right click while dragging your mouse to adjust viewing angle



PCB Board shape

1. Be on Mechanical Layer 1
2. Select Rectangle tool
3. Draw a rectangle where you want your board to end
4. Design > Board shape > Define board shape from selected objects



Review Questions

Questions are from sessions 1 and 2. All answers are on this presentation

- I accidentally hit 'x' on my properties, components, and projects viewer. How do I make them visible again?
- I need to integrate a sensor into my PCB. Where on the sensor's datasheet can I find one or more ways to wire its pins?
- I just finished a schematic and immediately tried to update the PCB document. Why did I get an error?
- How do you rotate a component's schematic symbol and how do you rotate its PCB footprint?
- I'm not sure if I imported an IC's 3D model along with its footprint. How do I check if the IC has a 3D model before updating the PCB document?

Next week's agenda:

- Wire PCB tracks and learn about their properties
- Electrical Rule Check (ERC) and Design Rule Check (DRC)
- Practice exporting PCB output files for a manufacturer
- Discuss voltage regulators and their working principles
- Q&A

Appendix

- Additional information

Troubleshooting

The screenshot displays the Altium Designer License Management window. The title bar reads "License Management - Altium Designer (26.10.1) - No License". The interface includes a sidebar with "Projects" and "Components and Tasks" sections. The main area is titled "License Management" and contains a user profile for Robert Bagalawis, a "Refresh" button, and a "Contact Admin" button. A large white text box is overlaid on the interface, stating: "If you see something like this, uninstall and reinstall altium". A modal dialog titled "License Renewal Failed" is open, showing an error icon and the message "Invalid client version" with an "OK" button. Below the dialog, a table lists licenses for "Altium Designer Professional" and "Altium Designer Viewer". The "Altium Designer Professional" license is marked as "Term-Based · N3VM-4B36" and has a "Roam" button. The "Altium Designer Viewer" license is marked as "On-Demand · Single Site · Viewer · YEFY-LKU9" and has a "Use License" button. A "Hide Expired Licenses" toggle is visible in the top right corner of the license list area.

License Management - Altium Designer (26.10.1) - No License

File View Project Window Help

Projects

Home Page License Management

License Management

License Management Extensions and Updates

Altium Account · Sign Out

Robert Bagalawis

Project Group 1.DsnWrk

Search

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Project Group 1.DsnWrk

License Management

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Invalid client version

License Renewal Failed

OK

Product

Altium Designer Professional

Product

Altium Designer Viewer

Subscription Renew

Seats

0/1

License Info

On-Demand · Single Site · Viewer · YEFY-LKU9

Term-Based · N3VM-4B36

Roam

Use License

Use License

Hide Expired Licenses

Refresh

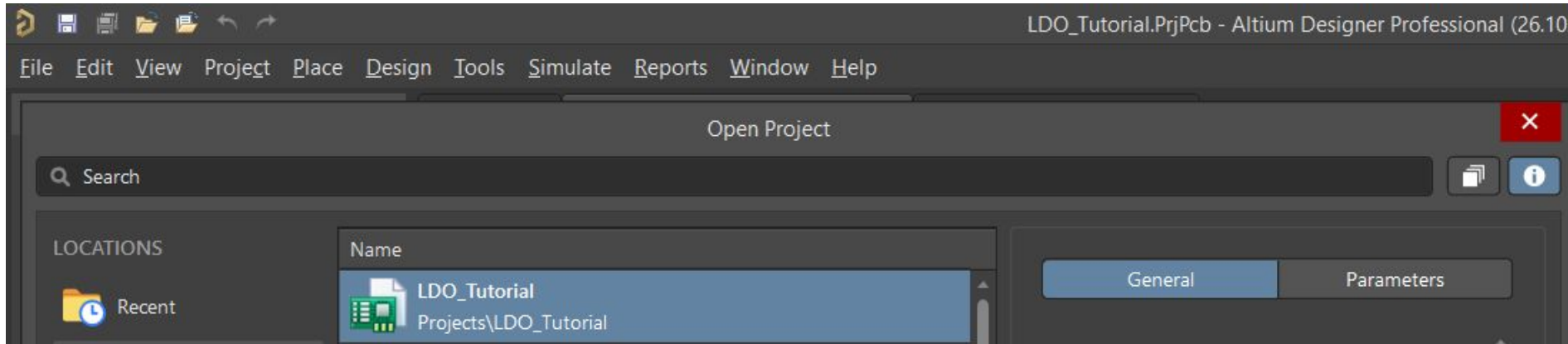
Contact Admin

Help Center

Opening Projects from Altium Application

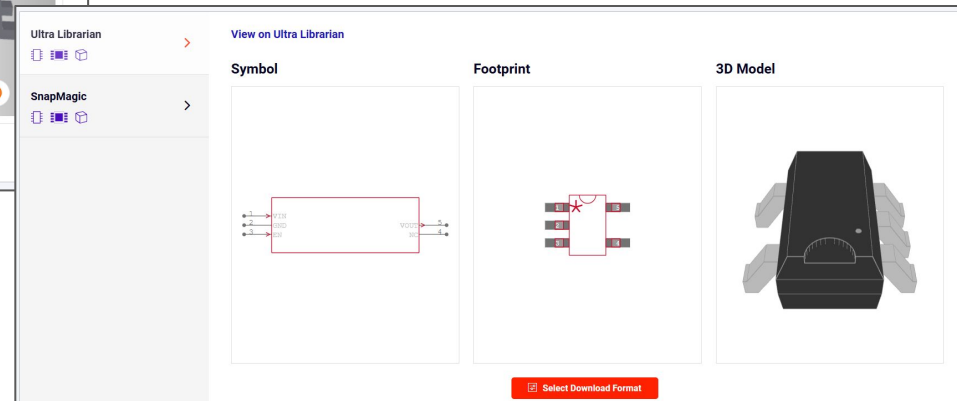
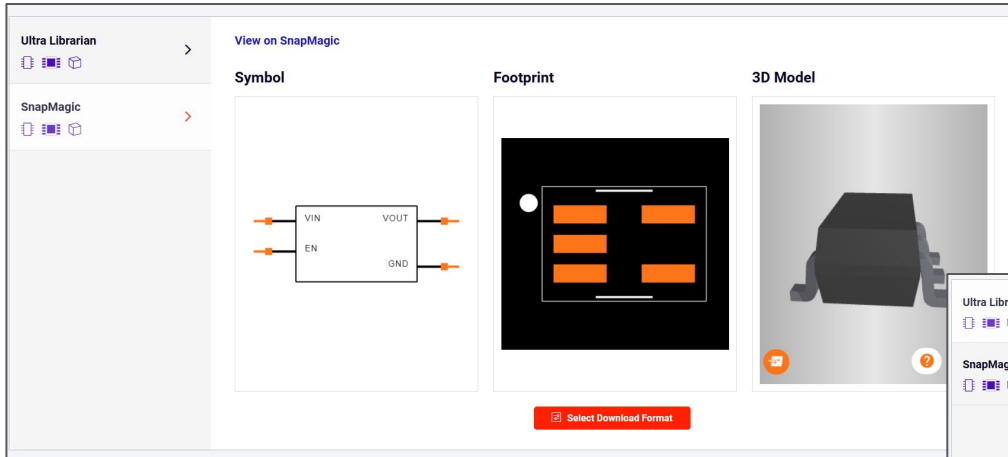
File > Open Project

This will menu all projects from your current workspace

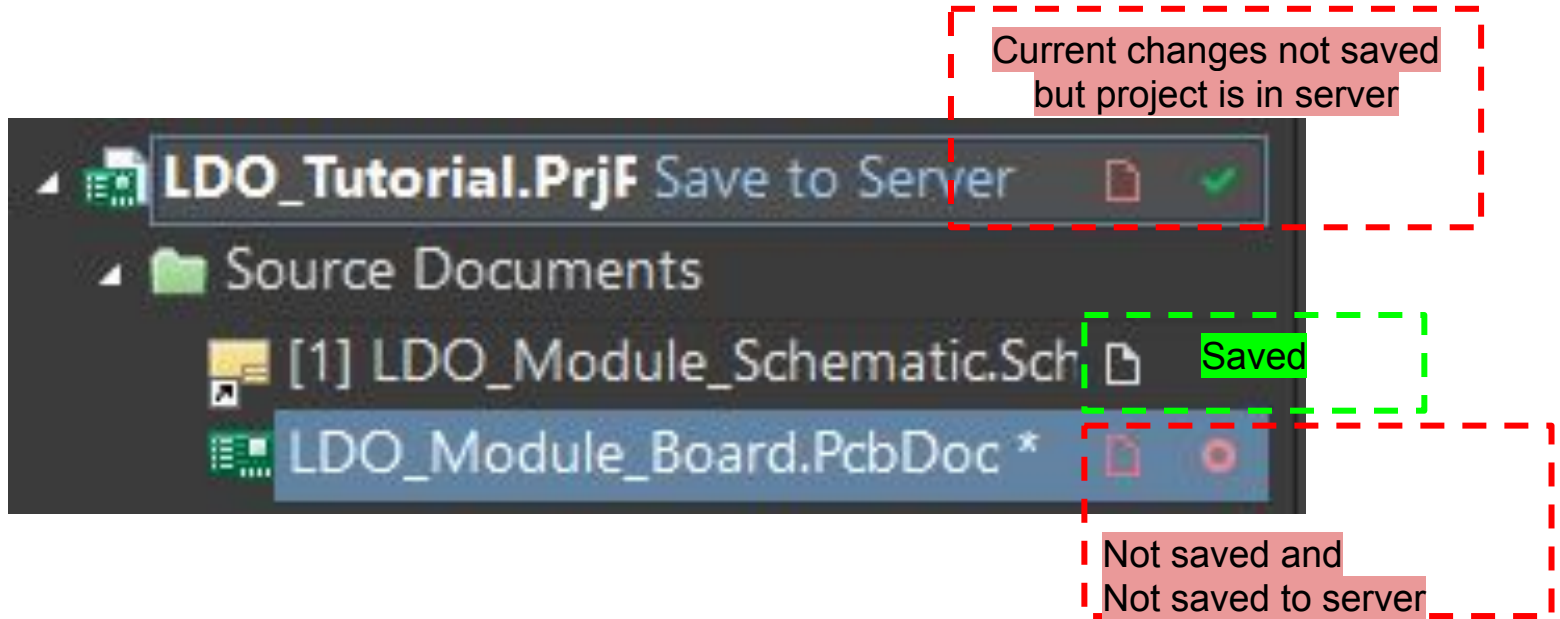


Same footprint, different symbol

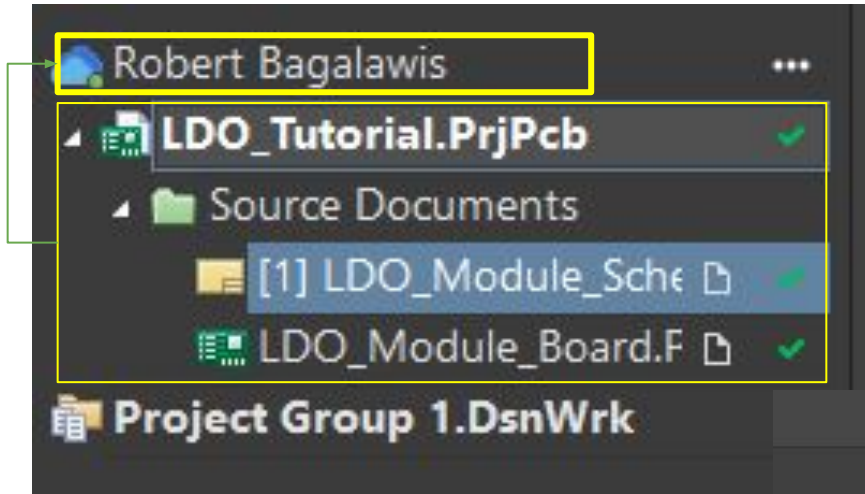
Some of you may have downloaded ECAD/3D models from Ultra Librarian while I downloaded from Snap Magic. The symbols are electrically the same. Snap Magic omitted the NC pin (No Connect)



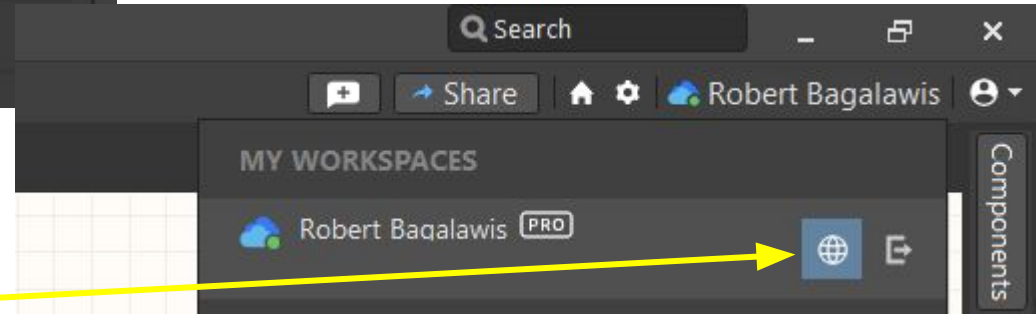
Save status



What saving to server does: Projects, files, and changes all get moved into current workspace

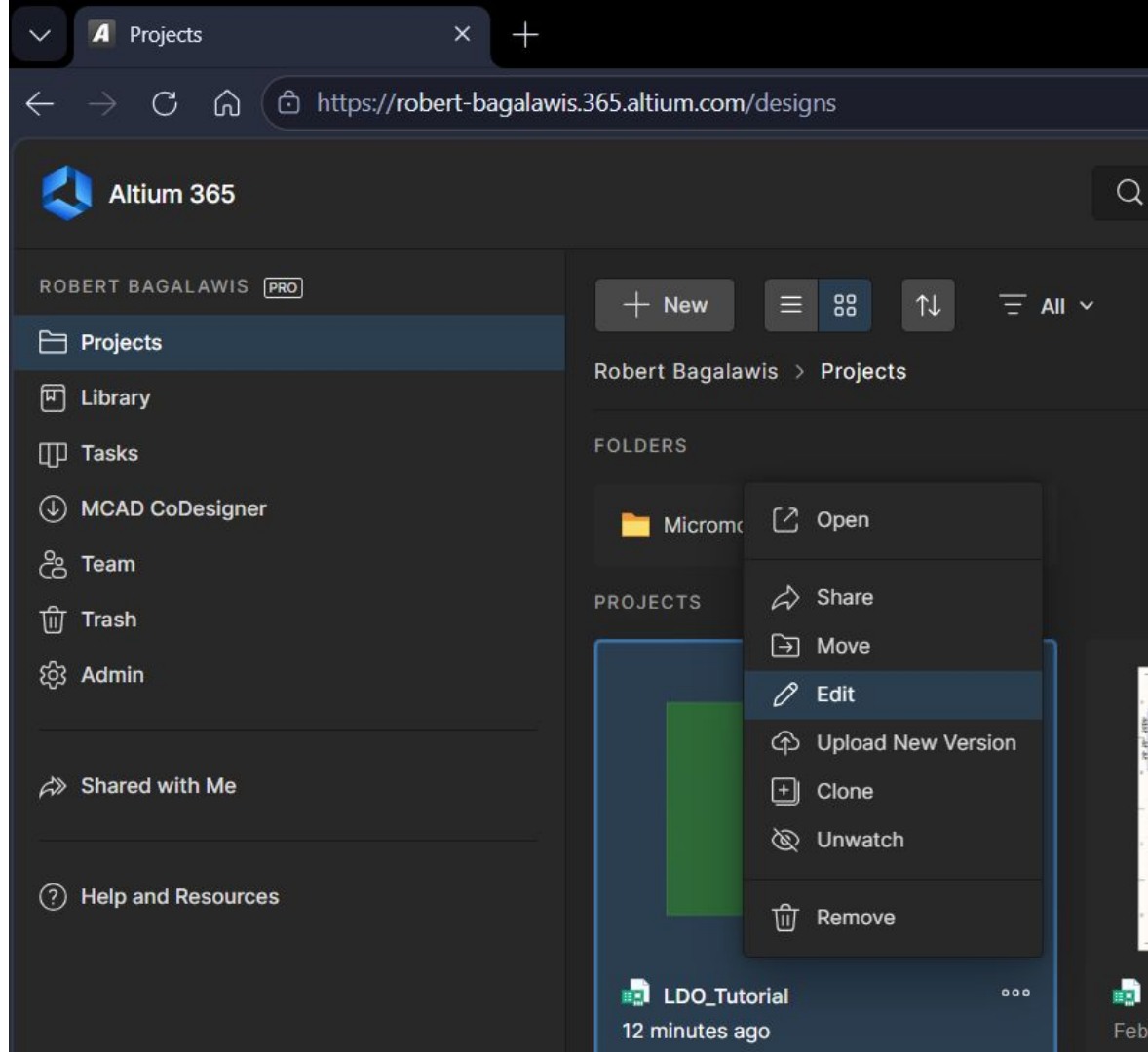


Check by opening your workspace on browser



Altium workspace on browser

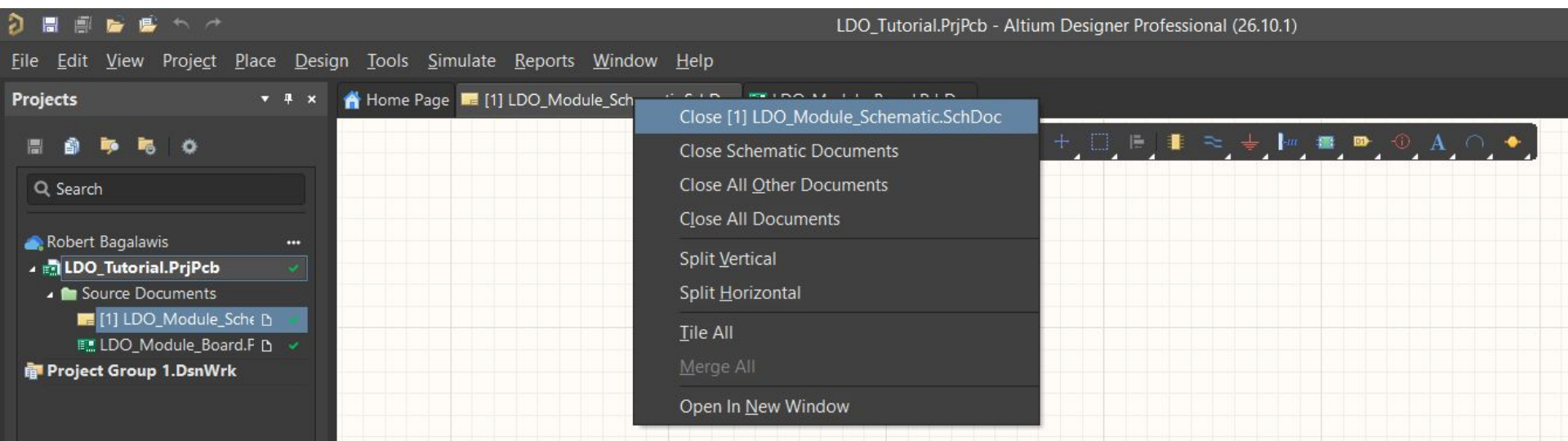
Only here, you can delete projects



Closing Documents

Right click on your current tab

You can close current doc, close all doc types, or all docs entirely



Closing a prj.pcb

After saving a project to server,

Right click on a prj.pcb > Close Project

Why?

- It saves space on your projects panel
- Reduces complication in project changes when collaborating
- Proper handoff when multiple people are working on a PCB
 1. User1 edits [name].proj.pcb > User1 saves changes to server > User1 closes [name].proj.pcb
 2. User2 opens [name].proj.pcb > User2 edits [name].proj.pcb > User2 saves changes to server > User2 closes [name].proj.pcb ...